#manydaughters: Pre-Analysis Plan for team

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1. Introduction

As part of the #manydaughters project (https://www.manydaughters.com), we aim to study the causal impact of daughters on gender equality (in terms of personal attitudes and household behavior) and political preferences. In this document we outline our analysis approach in greater detail, i.e., (pre-determined) hypotheses, data description, and our planned analysis approach. Importantly, at the time of writing this pre-analysis plan we **did not** have access to the data. Therefore, description of variables and the dataset are purely based on the recommended resources of the project-coordinators (i.e., SOEPcompanion, paneldata.org, and the DIW website).

2. Hypotheses

Consequently, we restate the pre-registered (directed) hypotheses from the call for participation:

H1*. Having daughters leads to stronger support for gender equality (as measured by attitudes).*
That is, denoting X as a measure of support for gender equality,

(X for respondents with daughters) > (X for respondents without daughters).

H2*. Having daughters leads to more left-leaning political preferences.*
That is, denoting Y as a measure of left-leaning political preferences,

(Y for respondents with daughters) > (Y for respondents without daughters).

H3*. Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).*
That is, denoting Z as a measure of gender equality in the household,

(Z for respondents with daughters) > (Z for respondents without daughters).

H4*. Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).*
That is, denoting X as a measure of support for gender equality,

X for male respondents with daughters − X for male respondents without daughters! > X for female respondents with daughters − X for female respondents without daughters!

3. Data

As indicated in the description of the project, we have access to the German Socio-Economic Panel's 2020 data release. In particular, we have access to the data between 2000 to 2020, including all individuals who completed the individual questionnaire in at least one of the years (n = 90,357). The following datasets are available to us: *pl, hl, biol, biobirth, bioparen, pgen, hgen, and ppathl*.

4. Data Analysis

In the following, we will outline our analysis approach in more detail. Based on prior literature on the effects of children's gender on gender equality attitudes (Borrell-Porta et al., 2019; Washington, 2008), we will estimate the following OLS baseline specification:[1]

$$y_i = \alpha + \beta_1 DAUGHTERS_i + \beta_2 NRCHILDREN_i + \sum_{k=3}^{K} \beta_k X_{ki} + \epsilon_i,$$

Where y_i is our dependent variable of unit *i*, $DAUGHTERS$ is the number of daughters, $NRCHILDREN$ is the total number of children (fixed effects; reference is zero), X is a vector of additional *k* control variables (outlined below), and ϵ are residuals. If not stated otherwise, we will make use of the *pl* dataset. The number of daughters and the number of children we retract from the *biobirth,* where we will merge this information using the personal identifier key.

Importantly, we will restrict our sample only to respondents (or households) with children. Furthermore, we follow significance guidelines proposed by Benjamin et al. (2018), meaning that *p*-values < 0.005 denote statistical significance (and *p*-values 0.05 suggestive evidence). Moreover, since the hypotheses are directed, we will use one-sided t-tests for finding positive effects (i.e., daughters leading to higher outcomes on average), which we base on one-sided t-tests on the residual degrees of freedom of the regressions. Consequently, our hypotheses are supported only when the one-sided t-test for a positive effect yields a p-value below 0.005.

Now we will focus on the specification for each hypothesis in particular.

H1: For H1, we will make use of one question in the 2018 wave, eliciting attitudes on gender equality. Specifically, we will take the following question:

Intro: I will read you a series of statements. To what degree do you personally agree with each statement? (1 - do not agree at all; 7 completely agree)

[1] For a discussion of identification assumptions, we refer to Washington (2008) and Borrel-Porta et al. (2019).

- *It's best if the man and the woman work the same amount so they can share the responsibility for taking care of the family and household equally.* (plh0308_v2)

Consequently, higher values suggest a stronger attitude towards gender equality. Inspired by relevant literature (see e.g., Oswald & Powdthavee, 2010), we include the following "standard" battery of control variables: gender (pla0009_h; reference being female), age (difference between survey year and birth year [ple0010_h] as a proxy) and age^2, monthly gross income (plc0013_h), type of employment (plb0022_h; reference being full-time employed), marital status (pld0131_h; reference being married), political party supported (plh0012_h, with reference being SPD), and highest obtained level of education (based on pgpbbil02 in *pgen*; dummy variable whether the individual obtained university degree in the past or not). Our coefficient of interest is β_1, indicating the average effect of one additional daughter on our gender equity attitudes. Finally, we will use heteroskedastic robust standard errors.

H2: In order to answer H2, we use the political affiliation item collected in the years 2005, 2009, 2014 and 2019 (i.e., plh0004; 0 - left aligned political preferences, 10 - right aligned political preferences). Importantly, to make coefficients intuitive, we will reverse code the scale, meaning that higher values indicate more left aligned political preferences. We will include the same battery of control variables as stated in H1, except that we don't include political party supported.[2] Additionally, since we have a longitudinal dataset here, we will include subject fixed effects, time (wave) fixed effects and cluster standard errors on the subject level. Again, β_1 is again our coefficient of interest.

H3: Since H3 is dedicated to behavior or outcomes on the household level, we will aggregate data of mothers and fathers,[3] and calculate the share of fathers' childcare on weekends (measured in hours/week and bi-annually from 2001 to 2020). We only use the questions addressing weekends (i.e., pli0019_h, pli0022) as childcare during the week might be constrained by other circumstances or obligations (e.g., employment).[4] In the set of control variables, we will include household income (hlc0005_h from *hl* dataset), and the employment status of mother and father (plb0022_h, reference again being full-time employment), respectively.[5] Furthermore, since we have longitudinal data, we will include household fixed effects, time (wave) fixed effects and cluster the standard errors on household level. Note that we still include the number of daughters in the household and the number of children overall, where β_1 is again our coefficient of interest.

[2] We don't include political parties supported because it is equivalent to our dependent variable of interest. However, in a robustness check, one could first divide parties into left and right wings and use this as an outcome variable (see e.g., Oswald & Powdthavee, 2010).
[3] Again, we will restrict the sample to people who have children. Additionally, we require them to live in the same household (using the household identifier).
[4] By contrast, weekends offer a more level playing field where both parents are more likely to be at home and available, thus providing a cleaner measure of how childcare responsibilities are shared within the household.
[5] Of course, it could be that employment status mediates the effect of having a daughter, i.e., fathers with daughter(s) decide to work less to take over childcare. However, although we consider this could theoretically be the case, we don't think this is very much realistic.

H4: For H4 we will run an identical model as outlined for H1, where we additionally include an interaction effect between DAUGHTERS and the male dummy. Consequently, the coefficient of this interaction is our variable of interest. If it is positive and significant it means that the effect of having daughters is significantly higher for men than for women (which constitutes our reference category for the gender dummy).

5. Robustness checks/further exploratory analysis paths

Given the richness of the dataset, we consider it worth mentioning that the research question(s) could also be examined with a variety of alternative model specifications.[6] One of such alternatives would be to make a difference-in-differences approach, leveraging the gender of the first-born child as a (quasi-) experimental setup. However, we do not consider that as our main analysis approach due to the fact that we don't have access to the data and as such do not have proxies for attrition rates over time and associated final sample sizes.[7]

References

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[6] In a typical publication process, we would deem it relevant to include alternative specifications as further robustness checks (preferably in a pre-registered exploratory analysis).

[7] As for H4, one could then include a three-way interaction (i.e., first-borne being a daughter x post birth x male), for which we would expect power issues to become relevant.